Interstellar Harvest: Growing Food Beyond Earth

ASSESSMENT RUBRIC · LDTC 615

Module 2 Assignment Rubric

System Design Brief: Your Hydroponic Grow Chamber

Total: 100 points | **CLOs:** 2, 3, 5 | **Length:** 300–400 words | **Type:** Analytic · Weighted criteria

CRITERIA	EXCELLENT 100–90%	PROFICIENT 89–75%	DEVELOPING 74–60%	BEGINNING BELOW 60%
Crop Selection Justification 25% of total	Names the selected crop and provides a specific, well-reasoned explanation of why it is suited for space agriculture. Goes beyond general statements — explains what properties (growth cycle length, edibility without processing, compact form, radiation tolerance, resource efficiency) make it a strong candidate for a spacecraft or habitat context.	Names the crop and gives a reasonable justification, but explanation stays somewhat general. Mentions space-relevant properties without connecting them directly to the constraints of the mission environment.	Names the crop but justification is vague or focused on Earth-based qualities rather than space-specific suitability. Does not demonstrate understanding of why space constraints matter for crop selection.	Crop is named without justification, or justification shows no understanding of why certain crops are more suitable for space agriculture than others.
System Configuration and Engineering Reasoning 35% of total	Describes the system configuration choices — lighting spectrum, nutrient delivery method, grow chamber setup — and explains the engineering reasoning behind each decision. Does not just list choices but explains why each choice was made in the context of the mission environment and the crop's needs. Demonstrates understanding that each design decision has tradeoffs.	Describes configuration choices adequately and offers reasoning for most of them. One or two decisions are stated without explaining the engineering rationale. Some tradeoffs are acknowledged but not fully developed.	Describes some configuration choices but reasoning is weak or absent. Reads more like a list of what was selected than an explanation of why those selections were made.	Configuration choices are absent, incorrect, or described without any engineering reasoning. Does not demonstrate understanding of how system design connects to plant growth outcomes.
Problem Identification and Iteration 25% of total	Identifies a specific problem encountered or limitation of the chosen configuration and describes a concrete change that would be made in the next iteration. Explanation is specific enough to be actionable — not just "I would improve the lighting" but what change would be made and why it would address the identified problem.	Identifies a problem or limitation and suggests a change, but the proposed fix is somewhat general. The connection between the identified problem and the proposed solution is present but not fully developed.	Mentions a problem but without specificity about what caused it or what would change. The iteration suggestion is vague or unconnected to the problem described.	No problem or limitation identified, or the response claims the system worked perfectly without acknowledging any constraints or tradeoffs inherent in any design.
Unanswered Question 10% of total	Poses a specific, genuine question that the simulation raised but did not answer. Points toward a real engineering or scientific challenge beyond what the simulation modeled. Shows awareness that the simulation is a simplified representation of a more complex real-world problem.	Question is relevant but somewhat broad or predictable. Points toward an interesting challenge without demonstrating deep engagement with the gap between simulation and reality.	Question is vague or restates something the simulation already addressed. Does not demonstrate awareness of what the simulation left unexplored.	No question posed, or question is unrelated to the simulation experience or Module 2 content.
Writing Clarity and Plain Language 5% of total	Writing is clear and accessible to a mission planning audience with no horticulture background, as the prompt instructs. Technical terms are explained when introduced. Stays within the 300–400 word range. Written in paragraph form.	Mostly clear with minor lapses into unexplained technical jargon. Stays close to the word range and paragraph format.	Writing is uneven. Some sections are clear, others assume horticulture knowledge the audience does not have. May be significantly over or under the word range.	Writing is unclear, heavily technical without explanation, or not in paragraph form. A mission planning audience without horticulture background could not follow the brief.

SCORE KEY ● Excellent: 90–100 pts ● Proficient: 75–89 pts ● Developing: 60–74 pts ● Beginning: Below 60 pts

EQUITY & ACCESSIBILITY NOTE

This rubric evaluates engineering reasoning and design thinking, not prior knowledge of hydroponics or plant science. A learner encountering these concepts for the first time in Module 2 who reasons carefully from the simulation experience and reading content will score as well as someone with an agricultural background. Plain language explanations of technical concepts receive full credit. The 300–400 word range is intentionally narrow to reduce barriers for learners who find open-ended writing constraints intimidating, while leaving room for learners who want to develop their reasoning more fully.